Lec#13: File Processing, part-I

What is a File?

A file is a sequence of data that is stored in secondary memory (disk drive).

Opening a File

We use the open function in Python to open a file. The open function creates a file object and associates it with a file on the disk.

<filevar> = **open** (<name>, <mode>)

Table 6-1 Some of the Python file modes

Mode	Description
'r'	Open a file for reading only. The file cannot be changed or written to.
'w'	Open a file for writing. If the file already exists, erase its contents. If it does not exist, create it.
'a'	Open a file to be written to. All data written to the file will be appended to its end. If the file does not exist, create it.

Reading Data From a File

- <filevar>.read() – returns the entire remaining contents of the file as a single (possibly large, multi-line) string. Watch out for final \n
- <filevar>.readline() – returns the next line of the file. This is all text up to and including the next newline character
- <filevar>.readlines() – returns a list of the remaining lines in the file. Each list item is a single line including the newline characters.
- File Processing Loop-Python treats the file itself as a sequence of lines!

```
infile = open(someFile, "r")
for line in infile:
    # process the line here
infile.close()
```

❖Learn more through interactive examples: https://runestone.academy/ns/books/published/thinkcspy/Files/ReadingaFile.html

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Writing Data in a File

If you open an existing file for writing, you wipe out the file's contents. If the named file does not exist, a new one is created.

```
outfile = open("mydata.out", "w")
outfile.write('Hello world\n, this is great')
```

May use `writelines()` for writing sequence (list) of strings

```
outfile = open("mydata.out", "w")
outfile.writelines(['Hello world\n', 'this is great'])
```